GLOBAL ELECTRONICS RETAILER

Business Intelligence & Power BI Project Documentation

End-to-end data audit, cleaning, validation, modeling, DAX development, interactive dashboard design, and business insights.

Project scope	Retail sales, profitability, customers, products, stores and currencies
Data preparation	Python / Pandas
Business intelligence	Microsoft Power BI
Dashboard pages	2
Primary analysis	Sales, profit, margin, channels, customers, products and trends

Prepared from the completed project implementation and supplied evidence.

1. Executive Summary

The Global Electronics Retailer project was developed as an end-to-end Business Intelligence solution to transform raw retail transaction data into an interactive Power BI dashboard for analyzing sales performance, profitability, customer behavior, product performance, and sales-channel effectiveness.

The project followed a structured workflow: **Data Audit** → **Data Cleaning** → **Validation** → **Data Modeling** → **DAX Development** → **Dashboard Design** → **Interactive Analysis** → **Business Insights**.

The final Power BI solution consists of two analytical pages: **Executive Sales & Profitability Overview** and **Product & Customer Insights**.

The dashboard enables users to evaluate business performance by year, country, sales channel, category, customer segment, product and other dimensions, while providing year-over-year comparisons and profitability analysis.

2. Project Objective

- Measure revenue, gross profit, gross margin, orders and units sold.
- Analyze sales and profit trends over time.
- Identify high-performing product categories and individual products.
- Compare Online and In-Store sales-channel performance.
- Analyze customer distribution, demographics and customer value.
- Identify top customers and top products by sales.
- Investigate factors associated with changes in Total Sales.

3. Business Problem

The retailer's data spans transactions, products, customers, stores, currencies, exchange rates and dates. Raw transactional data makes it difficult for management to identify trends and performance differences.

The BI solution therefore consolidates these datasets into a reliable analytical model and provides an interactive interface for executive monitoring, profitability analysis, product analysis, customer segmentation, geographic analysis and channel comparison.

4. Dataset Overview

Dataset	Rows	Columns	Purpose
Sales	62,884	9	Transaction-level sales
Products	2,517	10	Product and pricing information
Customers	15,266	10	Customer information
Stores	67	5	Store/location information
Exchange Rates	11,215	3	Currency exchange rates
Data Dictionary	37	3	Field definitions

The Sales dataset represents the central transactional component of the model.

5. Data Audit

Before building the Power BI model, the datasets were systematically audited using Python. The audit covered dataset dimensions, data types, missing values, duplicates, primary-key integrity, foreign-key integrity, transaction uniqueness, date validity, numeric fields, categorical values, currency consistency, exchange-rate coverage and business-rule

anomalies.

This process was used to prevent data-quality issues from being carried into the reporting layer.

6. Data Quality Findings

6.1 Missing Delivery Dates

There were **49,719 missing Delivery Dates (79.06%)** in Sales. Investigation showed that this pattern was systematic rather than random.

Channel	Transaction Lines	Missing Delivery Date
In-Store	49,719	100%
Online	13,165	0%

The missing dates were therefore interpreted as an expected characteristic of In-Store transactions. The records were not removed.

6.2 Product Price Formatting

Product cost and price values contained currency formatting such as **$6.62**, **$960.82** and **$2,899.99**. The cleaning process removed currency symbols and comma separators and converted the fields to numeric values.

Final validation confirmed **0 invalid Unit Cost values** and **0 invalid Unit Price values**.

6.3 Customer State Code

Ten customers had missing State Code values. Investigation showed that their State field was populated as Napoli, Italy. The missing codes were retained as a limited data-quality exception rather than removing the customers.

6.4 Store Square Meters

One missing Square Meters value belonged to StoreKey = 0, the Online channel. Because this is not a physical store, square-meter information is not applicable and the null was retained.

7. Duplicate and Key Validation

The audit confirmed zero exact duplicates across Sales, Products, Customers, Stores and Exchange Rates.

Dimension key validation confirmed no duplicate or null primary keys for CustomerKey, ProductKey or StoreKey. Sales foreign-key validation found zero unmatched CustomerKey, ProductKey or StoreKey values.

The transaction-level key **Order Number + Line Item** was also validated with zero duplicate combinations.

8. Date Validation

Check	Result
Order Date range	1 Jan 2016 – 20 Feb 2021
Invalid Order Dates	0
Delivery Date range	6 Jan 2016 – 27 Feb 2021
Missing Delivery Dates	49,719
Delivery before Order	0
Minimum delivery duration	1 day
Maximum delivery duration	17 days

Check	Result
Average delivery duration	4.53 days
Median delivery duration	4 days

9. Sales Channel Structure

Channel	Transaction Lines	Orders	Customers	Units
In-Store	49,719	20,746	10,657	156,446
Online	13,165	5,580	4,547	41,311

StoreKey = 0 was validated as the intentional representation of the Online channel and was not treated as an unmatched store key.

10. Currency and Exchange Rate Validation

Sales contain five currencies: USD, EUR, GBP, CAD and AUD. The Exchange Rates dataset contains the same five currencies.

Each currency contains 2,243 dates, with one exchange rate per currency/date combination. The exchange-rate date range is 1 January 2015 to 20 February 2021, with no invalid exchange-rate dates identified.

11. Final Data Validation

Validation Area	Final Result
Sales row count	62,884
Products row count	2,517
Customers row count	15,266
Stores row count	67
Exchange Rates row count	11,215
Invalid product cost values	0
Invalid product price values	0
Invalid order dates	0
Duplicate records	0
Duplicate dimension keys	0
Unmatched foreign keys	0
Duplicate Order Number + Line Item	0

Expected nulls were documented rather than incorrectly removed. The final datasets were suitable for Power BI modeling.

12. Power BI Data Model

The final Power BI model contains six tables:

- Sales_Cleaned
- Products_Cleaned

- Customers_Cleaned
- Stores_Cleaned
- Exchange_Rates_Cleaned
- Date

Sales_Cleaned functions as the central transaction table, while the other tables provide descriptive and analytical dimensions. The dedicated Date table supports time-based analysis and year-over-year calculations.

13. Date Dimension

- Date
- Month
- Month Number
- Month Short
- Quarter
- Year
- Year Month
- Year Month Label
- Year Month Sort

The additional sort fields support chronological presentation of months and year-month values rather than alphabetical sorting.

14. DAX Development

Sales & Profitability

Total Sales; Total Profit; Total Cost; Total Quantity; Total Profit %; Profit Margin %; In-Store Margin %; Online Margin %; Margin Difference.

Order & Customer Metrics

Total Orders; Total Customers; Average Order Value; Average Sales per Customer; Customers % of Total.

Year-over-Year Analysis

Total Sales PY; Total Sales vs PY; Total Cost PY; Total Cost vs PY; Total Profit PY; Total Profit vs PY; Total Orders PY; Total Orders vs PY; Total Quantity PY; Total Quantity vs PY; Profit Margin PY; Profit Margin vs PY; Total Customer PY; Total Customer vs PY.

Customer Segmentation

Customer Age; Age Group; Age Group Sort.

Display Measures

Display-oriented measures include Total Sales PY Display, Total Profit PY Display, Total Cost PY Display, Total Orders PY Display, Total Quantity PY Display, Profit Margin PY Display, Total Sales % Display and Total Profit % Display.

The report intentionally does not reproduce underlying DAX formulas where those formulas were not supplied in the project evidence.

15. Dashboard Architecture

The final solution contains two pages.

Page 1 — Executive Sales & Profitability Overview

This page is designed for executive-level monitoring of revenue, profitability, trends, category performance and channel effectiveness.

- Five executive KPI cards: Total Sales, Gross Profit, Gross Margin %, Total Orders and Units Sold.
- Monthly Sales & Profit Trend using combined columns and a line.
- Sales & Profit by Category comparison.
- Online vs In-Store Performance comparison.
- Category Profitability comparing In-Store Margin %, Online Margin % and Margin Difference.
- Key Insights panel summarizing major findings.

Page 2 — Product & Customer Insights

This page focuses on customer scale, demographics, geography, customer value, product rankings and sales drivers.

- Total Customers, Total Orders, Units Sold, Average Order Value and Average Sales per Customer.
- Sales Contribution by Age Group & Gender.
- Customers by Country with customer share.
- Key Influencers analysis for Total Sales.
- Top 10 Customers by Sales.
- Top 10 Products by Sales.
- Key Insights panel.

16. Dashboard Interactivity

The screen recording demonstrates that the dashboard is interactive rather than static. Year and Country slicers change the analytical context and dynamically update KPI cards and dependent visuals.

The interaction flow supports: **Overall performance** → **specific year** → **specific country** → **detailed product/customer analysis.**

The recording also demonstrates interaction with Power BI analytical visuals, including the Key Influencers analysis.

17. Business Insights

17.1 Overall profitability

The overall dashboard view reports approximately **$55.76M Total Sales**, **$32.66M Gross Profit** and **58.6% Gross Margin**.

17.2 In-Store channel dominance

In-Store contributes approximately **79.6% of total sales**, while Online contributes approximately 20.4%. Online and In-Store margins remain broadly comparable in the displayed analysis.

17.3 Strong product categories

The dashboard identifies **Computers** and **Home Appliances** among the strongest profit-generating categories.

17.4 Geographic customer concentration

The United States represents approximately **48% of total customers** in the overall customer view.

17.5 Demographic differences

The Age Group & Gender analysis highlights substantial sales contribution from older customer segments. The dashboard specifically identifies the 65+ group as having a notable gender sales gap, with male customers contributing approximately **$559K more** than female customers in the displayed analysis.

17.6 Desktop sales driver

The Key Influencers analysis identifies **Desktops** as a strong sales driver and indicates an approximately **$38K increase in average Total Sales** associated with the identified condition.

18. Business Recommendations

- **Maintain focus on high-performing categories:** Continue monitoring Computers and Home Appliances for inventory, promotional and assortment decisions.
- **Strengthen Online-channel growth:** Since Online maintains broadly comparable margins while representing a smaller sales share, digital promotions, bundles and targeted acquisition could be evaluated.
- **Protect high-value customers:** Use the Top 10 Customers analysis to support retention, loyalty and personalized offers.
- **Use demographic segmentation:** Evaluate marketing and product strategies by age and gender rather than applying a single strategy to all customers.
- **Monitor geographic concentration:** With the United States representing a large share of customers, track expansion opportunities in other established markets.

19. Dashboard Design Approach

The dashboard uses a consistent executive-oriented visual design with a dark professional background, high-contrast KPI cards, blue-based visual language, bright accent colors, structured analytical sections, clear information hierarchy, interactive slicers and dedicated Key Insights panels.

The design prioritizes rapid executive consumption while retaining sufficient detail for deeper analysis.

20. Visualization Selection Rationale

Visual	Purpose
KPI Cards	Immediate performance monitoring
Line/Column Chart	Sales and profit trend analysis
Horizontal Bar Chart	Category/product ranking
Comparison Table	Channel performance comparison
Conditional bars	Margin comparison
Age/Gender Chart	Demographic comparison
Country Table/Bar	Geographic customer distribution
Top 10 Charts	Ranking high-value entities
Key Influencers	Driver exploration
Slicers	Interactive filtering

21. Technical Implementation Summary

Area	Implementation
Data preparation	Python / Pandas

Area	Implementation
Data audit	Profiling, missing values, duplicates, keys, dates and business rules
Data cleaning	Currency-format removal and numeric conversion
Business Intelligence	Microsoft Power BI
Modeling	Fact/dimension-oriented relational model
Calculations	DAX measures and calculated columns
Analytics	Descriptive, comparative, time-series, profitability, customer, product, geographic and driver analysis

22. Data Limitations and Considerations

- **Delivery Date:** Unavailable for all In-Store transactions; delivery-time analysis should therefore be restricted to meaningful delivery records.
- **Customer State Code:** Ten customers have missing State Code values while State information remains available.
- **Online Store Area:** StoreKey 0 represents the Online channel and therefore has no physical square-meter measurement.
- **Filter Context:** KPI values and some displayed insights depend on active Year and Country selections.

23. Quality Assurance

The project incorporated multiple validation layers before dashboard development.

- **Structural integrity:** no duplicate records, no duplicate dimension keys and no unmatched foreign keys.
- **Numerical integrity:** product costs and prices successfully converted to numeric with zero invalid final values.
- **Temporal integrity:** valid Order Dates, valid Delivery Dates where present, and no delivery-before-order anomalies.
- **Business-rule integrity:** Online StoreKey and expected In-Store delivery-date nulls correctly interpreted.

These checks reduce the risk of reporting incorrect business results caused by underlying data-quality problems.

24. Conclusion

The Global Electronics Retailer project demonstrates a complete BI workflow from raw data quality assessment through interactive business reporting.

The underlying datasets were audited, investigated, cleaned and validated before being introduced into Power BI. The final model organizes the data into a structured analytical environment supported by DAX measures and a dedicated Date table.

Page 1 focuses on overall sales, profitability, category performance, trends and channel comparison. Page 2 focuses on customers, demographics, geographic distribution, top customers, top products and sales drivers.

The resulting solution provides management with an interactive mechanism to monitor performance, investigate business drivers, compare channels and identify opportunities across products and customer segments.

Project flow: Raw Retail Data → Python Data Audit → Cleaning & Transformation → Data Validation & QA → Power BI Data Model → DAX Measures & Calculated Fields → Interactive Dashboard → Business Insights & Recommendations